

AYUSH KASHYAP

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Summary

Backend and data engineer who builds things because they are interesting, not because they are assigned. Production experience with **Python**, **Go**, **PostgreSQL**, **distributed systems**, and **large-scale ETL pipelines**. Active open source contributor to **Ray** and **Milvus** with **7+ merged PRs** including feature-level C++ database engine work. Strong data quality instincts, comfortable with ambiguity, and genuinely enjoys digging into how systems work at a low level.

Experience

Software Engineer · Constructure AI

Dec 2025 – Apr 2026 · Remote

- Built **ETL pipelines** in Python processing **500MB+ BIM/IFC datasets** end-to-end with ingestion, schema mapping, transformation, and loading into **PostgreSQL + pgvector**; implemented **data quality checks**, automated retries, and **SQL-based validation** (joins, aggregations, indexing) at each stage.
- Architected **RESTful microservices** with telemetry and observability instrumentation, reducing **p95 latency from 1.8s to under 600ms** via **Redis caching** and async job queues; monitored across test, integration, and production with structured logging and alerting.
- Built a **multi-agent orchestration system** in Python coordinating **4+ LLM agents** across a **DAG pipeline** with automated validation and quality checks at each agent boundary, reducing error propagation by **40%**.
- Implemented a **hybrid retrieval system** combining **pgvector semantic search** with structured **SQL filtering**; applied **query optimization and indexing strategies** for concurrent low-latency access, improving precision by **28%**.

Open Source Contributions

Milvus (milvus-io/milvus)

5+ merged PRs · [PR #51051](#) · [PR #50666](#) · [PR #51636 \(open\)](#)

- Implemented a new **Decimal scalar field type** end-to-end in a production vector database: protobuf, **Go** parser, **C++ segcore** storage/indexing/query execution — **1000+ line size/XXL PR**, full CI passing; extended the filter expression engine with bitwise operators (**500+ lines across 8 files**); implementing **nested arithmetic operations** in scalar filters ([issue #51269](#)).

Ray (ray-project/ray)

3+ merged PRs · [PR #63952](#) · [PR #64546](#)

- Patched a **critical crash loop** in Ray Core's C++ OOM killer; contributed to **KV-cache-aware LLM request routing** ([PR #64546](#)) — navigated a 500K+ line codebase, diagnosed the root cause from logs, and shipped with cross-boundary unit tests.

Projects

Vigil — Autonomous PostgreSQL Observability and Auto-Remediation

[GitHub](#)

- Built a **Go collector** that polls **pg_stat_statements**, **pg_stat_activity**, and table/index stats to detect 7 real database problems (missing indexes, unused indexes, N+1 patterns, correlated subqueries, unbounded queries, idle transactions, connection exhaustion); persists findings to **SQLite** in WAL mode for concurrent reads.
- Built a **Python brain** that generates fixes — deterministic for index changes and N+1 batching, **LLM-proposed** (Groq) for query rewrites — and validates every fix in a disposable **pg_dump sandbox**: benchmarks before/after, diffs result sets for LLM rewrites, and silently discards anything that doesn't clear a **20% improvement threshold**; caught 6 subtle LLM rewrites returning wrong results before they were ever considered.
- Diagnosed and fixed a systemic **pg_stat_statements scoping bug** midway through development: the collector's polling query lacked a **WHERE dbid** filter, causing 4,710 orphaned rows from dropped sandbox clones to silently corrupt detection logic — root-caused from first principles.

GoChess Engine

[Demo](#)

- Built a fully playable chess engine in **Go** implementing **Minimax Search** with **Alpha-Beta Pruning** that reduces the effective game-tree by **60%** and completes depth-5 analysis in **under 200ms**; cross-compiled to **WebAssembly (WASM)** and offloaded search to **Web Workers**, eliminating UI blocking and achieving **2–3×** throughput over a single-threaded baseline.
- Integrated **Zobrist Hashing** with a **Transposition Table** caching **100K+ positions/session** to avoid re-evaluating repeated board states; added **Quiescence Search** to prevent the horizon effect in tactically volatile positions and a **Piece-Square Table (PST)** evaluation function for positional understanding — each addition benchmarked against known positions for correctness.

Technical Skills

Go, Python, TypeScript, JavaScript, C++, Java, PostgreSQL, Redis, ETL Pipelines, Distributed Systems, SQL Optimization, Docker, CI/CD, Git, Ray, LangChain, LangGraph, RAG, pgvector, Milvus, Automated Testing, Telemetry, Linux

Education

Punjab Engineering College (PEC), Chandigarh

B.Tech in Mechanical Engineering

2023 – 2027

CGPA: 8.23 / 10